03c CFPB Time Series Deep Learning (NeuralProphet)

Objective: Build NeuralProphet baselines for quarterly Credit Reporting complaints using the same y and 33/10/10 split framing from 03a/03b, with 4-quarter-ahead evaluation.

In [2]:
```
# ============================================================================
# IMPORTS
# ============================================================================
import os
import time
import warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from neuralprophet import NeuralProphet

warnings.filterwarnings("ignore")
```

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```

In [3]:
```
# ============================================================================
# DATA SETUP (MATCH 03a/03b)
# ============================================================================
cols = ["Product", "year_quarter", "Date received"]
df = pd.read_parquet("../data/processed/full_dataset.parquet", columns=cols)

if "year_quarter" not in df.columns:
    df["Date received"] = pd.to_datetime(df["Date received"], errors="coerce")
    df["year_quarter"] = df["Date received"].dt.to_period("Q").astype(str)

mask_cr = df["Product"].str.contains("Credit reporting", case=False, na=False)
series_df = (
    df.loc[mask_cr]
      .groupby("year_quarter")
      .size()
      .rename("y")
      .to_frame()
      .sort_index()
)
series_df.index = pd.PeriodIndex(series_df.index, freq="Q")
series_df = series_df.loc[:"2025Q4"].copy()
y = series_df["y"].astype(float)

# Same split logic as 03a/03b target split (expected 33/10/10 for this y window).
n = len(y)
test_size = max(4, n // 5)
val_size = test_size
while n - (val_size + test_size) < 8 and test_size > 2:
```

```
    test_size -= 1
    val_size = test_size

train = y.iloc[: n - (val_size + test_size)]
val = y.iloc[n - (val_size + test_size) : n - test_size]
test = y.iloc[n - test_size :]

print(f"Points: {len(y)} | train={len(train)}, val={len(val)}, test={len(test)}")
print(f"Window: {y.index.min()} -> {y.index.max()}")
```

```
Points: 53 | train=33, val=10, test=10
Window: 2012Q4 -> 2025Q4
```

In [ ]:

```
# ============================================================================
# SHARED EVAL HELPERS
# ============================================================================
def smape(y_true, y_pred):
    y_true = np.asarray(y_true, dtype=float)
    y_pred = np.asarray(y_pred, dtype=float)
    denom = np.abs(y_true) + np.abs(y_pred)
    ratio = np.where(denom == 0, 0.0, 2.0 * np.abs(y_true - y_pred) / denom)
    return float(100.0 * np.mean(ratio))

def mae(y_true, y_pred):
    return float(np.mean(np.abs(np.asarray(y_true) - np.asarray(y_pred))))

def rmse(y_true, y_pred):
    return float(np.sqrt(np.mean((np.asarray(y_true) - np.asarray(y_pred)) ** 2)))

y_val_multi = None
y_test_multi = None

def append_result(model_name, val_pred, test_pred, runtime_sec):
    if np.ndim(val_pred) == 2:
        if y_val_multi is None or y_test_multi is None:
            raise ValueError("y_val_multi and y_test_multi must be built before mul
        v_scores = [
            (
                smape(y_val_multi.iloc[:, h].values, val_pred[:, h]),
                mae(y_val_multi.iloc[:, h].values, val_pred[:, h]),
                rmse(y_val_multi.iloc[:, h].values, val_pred[:, h])
            )
            for h in range(val_pred.shape[1])
        ]
        t_scores = [
            (
                smape(y_test_multi.iloc[:, h].values, test_pred[:, h]),
                mae(y_test_multi.iloc[:, h].values, test_pred[:, h]),
                rmse(y_test_multi.iloc[:, h].values, test_pred[:, h])
            )
            for h in range(test_pred.shape[1])
        ]
        smape_val, mae_val, rmse_val = np.mean(v_scores, axis=0)
        smape_test, mae_test, rmse_test = np.mean(t_scores, axis=0)
    else:
        smape_val = smape(val.values, val_pred)
```

```
        mae_val = mae(val.values, val_pred)
        rmse_val = rmse(val.values, val_pred)
        smape_test = smape(test.values, test_pred)
        mae_test = mae(test.values, test_pred)
        rmse_test = rmse(test.values, test_pred)

    eval_rows.append({
        "Model": model_name,
        "sMAPE_val": smape_val,
        "MAE_val": mae_val,
        "RMSE_val": rmse_val,
        "sMAPE_test": smape_test,
        "MAE_test": mae_test,
        "RMSE_test": rmse_test,
        "runtime_sec": runtime_sec
    })

def plot_section_predictions(section_title, pred_map):
    fig, ax = plt.subplots(figsize=(10, 4))
    y_train_ts = pd.Series(train.values, index=train.index.to_timestamp(how="end"))
    val_idx = val.index.to_timestamp(how="end")
    test_idx = test.index.to_timestamp(how="end")

    ax.plot(y_train_ts.index, y_train_ts.values, color="steelblue", label="Observed
    ax.plot(val_idx, val.values, color="steelblue", linestyle="--", alpha=0.8, labe
    ax.plot(test_idx, test.values, color="steelblue", linestyle=":", alpha=0.9, lab

    for model_name, preds in pred_map.items():
        val_pred_h1, test_pred_h1 = preds
        ax.plot(val_idx, val_pred_h1, linestyle="--", marker="o", label=f"{model_na
        ax.plot(test_idx, test_pred_h1, marker="o", label=f"{model_name} - Test (h=

    ax.set_title(section_title)
    ax.legend(ncol=2, fontsize=8)
    plt.tight_layout()
    plt.show()

def to_np_df(s):
    return pd.DataFrame({"ds": s.index.to_timestamp(how="start"), "y": s.values})

def recursive_h1_with_actuals(model, history_df, actual_series):
    hist = history_df.copy()
    preds = []
    for ts, y_true in zip(actual_series.index.to_timestamp(how="start"), actual_ser
        fut = model.make_future_dataframe(hist, periods=1, n_historic_predictions=F
        fc = model.predict(fut)
        yhat = float(fc["yhat1"].dropna().iloc[-1])
        preds.append(yhat)
        hist = pd.concat([hist, pd.DataFrame({"ds": [ts], "y": [y_true]})], ignore_
    return np.asarray(preds, dtype=float)

def recursive_h_forecast_single(model, origin_history_df, h=4):
    hist = origin_history_df.copy()
    out = []
    for _ in range(h):
        fut = model.make_future_dataframe(hist, periods=1, n_historic_predictions=F
```

```
        fc = model.predict(fut)
        yhat = float(fc["yhat1"].dropna().iloc[-1])
        out.append(yhat)
        next_ds = (pd.Period(hist["ds"].iloc[-1], freq="Q") + 1).to_timestamp(how="
        hist = pd.concat([hist, pd.DataFrame({"ds": [next_ds], "y": [yhat]})], igno
    return np.asarray(out, dtype=float)

def direct_h_forecast_multi(model, origin_history_df, h=4):
    fut = model.make_future_dataframe(origin_history_df, periods=h, n_historic_pred
    fc = model.predict(fut)
    cols = [f"yhat{i}" for i in range(1, h + 1)]
    row = None
    for i in range(len(fc) - 1, -1, -1):
        cand = fc.iloc[i][cols]
        if cand.notna().all():
            row = cand
            break
    if row is None:
        row = fc[cols].ffill().iloc[-1]
    return row.values.astype(float)

def build_truth_matrix(split_series, h=4, n_origins=4):
    rows = []
    for i in range(n_origins):
        rows.append([float(split_series.iloc[i + j]) for j in range(h)])
    df_out = pd.DataFrame(rows, columns=[f"y_h{k}" for k in range(1, h + 1)])
    return df_out

eval_rows = []
np_h1_rows = []
```

In [5]:

```
# ============================================================================
# BUILD NP DATAFRAMES AND H=4 TRUTH TABLES
# ============================================================================
df_train = to_np_df(train)
df_val = to_np_df(val)
df_test = to_np_df(test)
df_train_val = to_np_df(pd.concat([train, val]))
df_all = to_np_df(y)

# For 4-quarter direct evaluation, use first 4 rolling origins in val and test wind
y_val_multi = build_truth_matrix(val, h=4, n_origins=4)
y_test_multi = build_truth_matrix(test, h=4, n_origins=4)

display(y_val_multi.head())
display(y_test_multi.head())
```

	y_h1	y_h2	y_h3	y_h4
0	68462.0	77872.0	80799.0	78278.0
1	77872.0	80799.0	78278.0	116089.0
2	80799.0	78278.0	116089.0	147089.0
3	78278.0	116089.0	147089.0	150013.0

	y_h1	y_h2	y_h3	y_h4
0	276130.0	287864.0	398641.0	516605.0
1	287864.0	398641.0	516605.0	661366.0
2	398641.0	516605.0	661366.0	775293.0
3	516605.0	661366.0	775293.0	1007034.0

1. Single-Step Baseline (AR + trend/seasonal)

In [ ]:
```
# ============================================================================
# SECTION 1: NP SINGLE-STEP
# ============================================================================
t0 = time.perf_counter()
m = NeuralProphet(
    n_forecasts=1,
    n_lags=6,
    weekly_seasonality=False,
    yearly_seasonality=True,
    learning_rate=0.1,
    ar_reg=0.5,
    trend_reg=0.1,
    trend_reg_threshold=True
)
_ = m.fit(df_train, freq="Q", validation_df=df_val, progress="off")
runtime_single = time.perf_counter() - t0

# h=1 walk-forward predictions over full val and test windows (length 10 each).
np_single_val_h1 = recursive_h1_with_actuals(m, df_train, val)
np_single_test_h1 = recursive_h1_with_actuals(m, df_train_val, test)

# h=1 metrics for comparison section.
np_h1_rows.append({
    "Model": "NP-Single (h=1)",
    "sMAPE_val": smape(val.values, np_single_val_h1),
    "MAE_val": mae(val.values, np_single_val_h1),
    "RMSE_val": rmse(val.values, np_single_val_h1),
    "sMAPE_test": smape(test.values, np_single_test_h1),
    "MAE_test": mae(test.values, np_single_test_h1),
    "RMSE_test": rmse(test.values, np_single_test_h1),
    "runtime_sec": runtime_single
})
```

```python
# h=4 recursive-from-single predictions for first 4 origins in val and test.
single_val_rows = []
for i in range(4):
    hist = to_np_df(pd.concat([train, val.iloc[:i]]))
    single_val_rows.append(recursive_h_forecast_single(m, hist, h=4))
np_single_val_h4 = np.vstack(single_val_rows)

single_test_rows = []
for i in range(4):
    hist = to_np_df(pd.concat([train, val, test.iloc[:i]]))
    single_test_rows.append(recursive_h_forecast_single(m, hist, h=4))
np_single_test_h4 = np.vstack(single_test_rows)

append_result("NP-Single", np_single_val_h4, np_single_test_h4, runtime_single)
print("NP-Single complete.")
```

```
INFO - (NP.config.__post_init__) - Note: Trend changepoint regularization is experim
ental.
WARNING - (NP.forecaster.fit) - When Global modeling with local normalization, metri
cs are displayed in normalized scale.
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.788% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.config.init_data_params) - Setting normalization to global as only one da
taframe provided for training.
INFO - (NP.utils.set_auto_seasonalities) - Disabling daily seasonality. Run NeuralPr
ophet with daily_seasonality=True to override this.
INFO - (NP.config.set_auto_batch_epoch) - Auto-set batch_size to 8
INFO - (NP.config.set_auto_batch_epoch) - Auto-set epochs to 430
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INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.788% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 79.412% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.143% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.778% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.378% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.947% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 76.923% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.5% of t
he data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.049% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.571% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 76.744% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.273% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.778% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.261% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.723% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-SUN corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-SUN corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 79.167% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-SAT corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-SAT corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 79.592% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 80.0% of t
he data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.431% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-MON corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-MON corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.846% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.788% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 79.412% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.143% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.778% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 79.412% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.143% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.778% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.378% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.143% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-WED corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.778% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.378% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
```

```
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.947% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.778% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.378% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 57.143% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.947% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 92D corresponds to 42.857% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 76.923% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
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ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 76.744% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
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INFO - (NP.df_utils._infer_frequency) - Major frequency 90D corresponds to 28.571% o
f the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
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or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.273% of
the data.
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
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INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.778% of
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or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.261% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 71.429% of
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INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 77.273% of
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INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
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Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
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Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.723% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-SUN corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
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or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
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Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
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Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
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WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
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or, BAS.
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 79.167% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-SAT corresponds to 42.85
7% of the data.
WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
ll be resampled according to given freq QS. Ignore                     message if ac
tual frequency is any of the following:  SM, BM, CBM, SMS, BMS, CBMS, BQ, BQS, BA,
or, BAS.
INFO - (NP.df_utils._infer_frequency) - Major frequency 13W-SAT corresponds to 42.85
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WARNING - (NP.df_utils._infer_frequency) - Dataframe has multiple frequencies. It wi
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INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
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INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
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the data.
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Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
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INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 79.592% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 85.714% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.data.processing._handle_missing_data) - Dropped 1 rows at the end with Na
Ns in 'y' column.
Predicting: |          | 0/? [00:00<?, ?it/s]
INFO - (NP.df_utils.return_df_in_original_format) - Returning df with no ID column
NP-Single complete.
```

2. Multi-Horizon Direct (n_forecasts=4)

In [11]:
```python
# ============================================================================
# SECTION 2: NP MULTI-HORIZON DIRECT
# ============================================================================
t0 = time.perf_counter()
m_multi = NeuralProphet(
    n_forecasts=4,
    n_lags=6,
    weekly_seasonality=False,
    yearly_seasonality=True,
    learning_rate=0.1,
    ar_reg=0.5,
    trend_reg=0.1,
    trend_reg_threshold=True
)
_ = m_multi.fit(df_train, freq="Q", validation_df=df_val, progress="off")
runtime_multi = time.perf_counter() - t0

# h=1 walk-forward (for section plot + comparison) using direct model's yhat1.
def h1_from_direct(model, history_df, actual_series):
    hist = history_df.copy()
    preds = []
    for ts, y_true in zip(actual_series.index.to_timestamp(how="start"), actual_ser
        vec = direct_h_forecast_multi(model, hist, h=4)
        preds.append(float(vec[0]))
        hist = pd.concat([hist, pd.DataFrame({"ds": [ts], "y": [y_true]})], ignore_
    return np.asarray(preds, dtype=float)

np_multi_val_h1 = h1_from_direct(m_multi, df_train, val)
np_multi_test_h1 = h1_from_direct(m_multi, df_train_val, test)

np_h1_rows.append({
    "Model": "NP-Multi (h=1)",
```

```
    "sMAPE_val": smape(val.values, np_multi_val_h1),
    "MAE_val": mae(val.values, np_multi_val_h1),
    "RMSE_val": rmse(val.values, np_multi_val_h1),
    "sMAPE_test": smape(test.values, np_multi_test_h1),
    "MAE_test": mae(test.values, np_multi_test_h1),
    "RMSE_test": rmse(test.values, np_multi_test_h1),
    "runtime_sec": runtime_multi
})

# h=4 direct predictions for first 4 origins in val and test.
multi_val_rows = []
for i in range(4):
    hist = to_np_df(pd.concat([train, val.iloc[:i]]))
    multi_val_rows.append(direct_h_forecast_multi(m_multi, hist, h=4))
np_multi_val_h4 = np.vstack(multi_val_rows)

multi_test_rows = []
for i in range(4):
    hist = to_np_df(pd.concat([train, val, test.iloc[:i]]))
    multi_test_rows.append(direct_h_forecast_multi(m_multi, hist, h=4))
np_multi_test_h4 = np.vstack(multi_test_rows)

append_result("NP-Multi", np_multi_val_h4, np_multi_test_h4, runtime_multi)
plot_section_predictions("NP Single vs Multi (h=1 shown)", {"NP-Single": (np_single

results = pd.DataFrame(eval_rows).copy()
for c in ["sMAPE_val", "MAE_val", "RMSE_val", "sMAPE_test", "MAE_test", "RMSE_test"
    results[c] = results[c].round(3)
display(results)
```

```
INFO - (NP.config.__post_init__) - Note: Trend changepoint regularization is experim
ental.
WARNING - (NP.forecaster.fit) - When Global modeling with local normalization, metri
cs are displayed in normalized scale.
INFO - (NP.df_utils._infer_frequency) - Major frequency QS corresponds to 78.788% of
the data.
INFO - (NP.df_utils._infer_frequency) - Defined frequency is equal to major frequenc
y - QS
INFO - (NP.config.init_data_params) - Setting normalization to global as only one da
taframe provided for training.
INFO - (NP.utils.set_auto_seasonalities) - Disabling daily seasonality. Run NeuralPr
ophet with daily_seasonality=True to override this.
INFO - (NP.config.set_auto_batch_epoch) - Auto-set batch_size to 8
INFO - (NP.config.set_auto_batch_epoch) - Auto-set epochs to 460
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The Kernel crashed while executing code in the current cell or a previous cell.

Please review the code in the cell(s) to identify a possible cause of the failure.

Click <a href='https://aka.ms/vscodeJupyterKernelCrash'>here</a> for more info.

View Jupyter <a href='command:jupyter.viewOutput'>log</a> for further details.
```

3. Time-Series Cross-Validation

In [ ]:

```
# ============================================================================
# SECTION 3: CV (NeuralProphet crossvalidation API varies by version)
# ============================================================================
cv_metrics_df = None
try:
    from neuralprophet import crossvalidation
    cv_out = crossvalidation(m_multi, df_all, freq="Q", fold_pct=0.1, fold_overlap_
    if isinstance(cv_out, (list, tuple)) and len(cv_out) > 1:
        cv_metrics_df = cv_out[1]
    elif isinstance(cv_out, pd.DataFrame):
        cv_metrics_df = cv_out
except Exception as e:
    print("crossvalidation() not available with current signature/version:", e)

if cv_metrics_df is not None:
```

```
    print("CV metrics (raw):")
    display(cv_metrics_df.head(20))
else:
    print("CV metrics unavailable via built-in API in this environment; proceed wit
```

4. Interpretability (Components + AR/Parameters)

In [ ]:
```
# ==============================================================================
# SECTION 4: INTERPRETABILITY
# ==============================================================================
forecast_for_components = m_multi.predict(m_multi.make_future_dataframe(df_train, p
_ = m_multi.plot_components(forecast_for_components)
try:
    _ = m_multi.plot_parameters()
except Exception as e:
    print("plot_parameters not available in this version:", e)

# Attempt AR weight extraction (version-dependent internals).
ar_weight_df = None
try:
    if hasattr(m_multi.model, "ar_weights"):
        w = m_multi.model.ar_weights.detach().cpu().numpy().reshape(-1)
        ar_weight_df = pd.DataFrame({"lag": np.arange(1, len(w) + 1), "weight": w})
        ar_weight_df["abs_weight"] = np.abs(ar_weight_df["weight"])
        ar_weight_df = ar_weight_df.sort_values("abs_weight", ascending=False).rese
except Exception as e:
    print("AR weight extraction unavailable in current NP internals:", e)

if ar_weight_df is not None:
    print("Top AR lags by |weight|:")
    display(ar_weight_df.head(12))
else:
    print("AR weight table not extracted; use component/parameter plots above.")
```

5. Compare vs 03b XGB/RF Baselines (h=1)

In [ ]:
```
# ==============================================================================
# SECTION 5: COMPARISON VS 03b TREE H1
# ==============================================================================
np_h1_df = pd.DataFrame(np_h1_rows).copy()
np_h1_df["Family"] = "neuralprophet-h1"

tree_h1_path = "../reports/2026-03-25/tables/03b_tree_h1_summary.csv"
if os.path.exists(tree_h1_path):
    tree_h1_df = pd.read_csv(tree_h1_path)
    combined_h1 = pd.concat([tree_h1_df, np_h1_df], ignore_index=True, sort=False)
else:
    print(f"Warning: {tree_h1_path} not found. Showing NeuralProphet h=1 only.")
    combined_h1 = np_h1_df.copy()

for c in ["sMAPE_val", "MAE_val", "RMSE_val", "sMAPE_test", "MAE_test", "RMSE_test"
    combined_h1[c] = pd.to_numeric(combined_h1[c], errors="coerce").round(3)
```

```
combined_h1 = combined_h1.sort_values(["sMAPE_test", "MAE_test", "runtime_sec"]).re
display(combined_h1)
```

6. Save Multi-Horizon Model Artifact for 04 South

In [ ]:
```
# ============================================================================
# SECTION 6: SAVE MODEL
# ============================================================================
import torch

model_dir = "../models"
os.makedirs(model_dir, exist_ok=True)

state_path = f"{model_dir}/03c_np_multi_state_dict.pt"
torch.save(m_multi.model.state_dict(), state_path)

# Save config and key metrics for reproducible reload setup.
meta_path = f"{model_dir}/03c_np_multi_meta.json"
pd.Series({
    "n_forecasts": 4,
    "n_lags": 6,
    "freq": "Q",
    "train_points": len(train),
    "val_points": len(val),
    "test_points": len(test)
}).to_json(meta_path, indent=2)

print("Saved NeuralProphet model state:", state_path)
print("Saved metadata:", meta_path)
```